Kalman Delta Networks 3:
Two-Axis Factorized Uncertainty

August 28, 2026

Abstract

Kalman Delta Networks treat a recurrent key–value memory as the posterior mean of a latent linear map. Existing variants retain uncertainty along only one axis. Diagonal KDN uses a key-space covariance shared by every value coordinate, $\boldsymbol{I}_{d_v} \otimes \boldsymbol{P}_t$, whereas a complementary value-wise isotropic model uses $\boldsymbol{B}_t \otimes \boldsymbol{I}_{d_k}$. This note first derives the value-wise model and its independent scalar information scans. It then introduces KDN-3, whose retained covariance is the two-axis separable family

$$\boldsymbol{\Sigma}_t = \boldsymbol{B}_t \otimes \boldsymbol{P}_t,$$

with both factors positive diagonal. The exact conditional mean has a simple rank-one form: a key-wise uncertainty direction multiplies value-wise uncertainty gates. The exact posterior covariance, however, is generally not another single Kronecker product. We derive the exact posterior, characterize the exceptional closure condition, and obtain the joint reverse-KL projection back to the separable family. That projection is a positive matrix-scaling problem, unique only up to an intrinsic factor gauge. Consequently, KDN-3 is a principled two-axis uncertainty model, but it does not inherit the independent Möbius scans or the single shared compact-WY memory system of one-axis KDN without an additional approximation. We also connect KDN-3 to Gated DeltaNet-2: both expose key- and value-axis modulation, but KDN-3 ties erasure and writing through one innovation, and their generic algebraic intersection is the scalar KDA rule.

1 Scope and notation

Let $K = d_k$ and $V = d_v$. The recurrent memory $\boldsymbol{S}_t \in \mathbb{R}^{K\times V}$ maps a key to a value. Its jth value column is $\boldsymbol{s}_{t,j} \in \mathbb{R}^K$, and column-major vectorization is

$$\boldsymbol{x}_t = \operatorname{vec}(\widetilde{\boldsymbol{S}}_t) = \begin{bmatrix} \widetilde{\boldsymbol{s}}_{t,1}^{\mathsf{T}} & \cdots & \widetilde{\boldsymbol{s}}_{t,V}^{\mathsf{T}} \end{bmatrix}^{\mathsf{T}}. \tag{1}$$

A tilde denotes the latent random memory, a bare symbol denotes its retained mean, and a hat denotes the exclusive one-step prediction before the current observation is assimilated. The observation model is

$$\boldsymbol{v}_t = \widetilde{\boldsymbol{S}}_t^{\mathsf{T}} \boldsymbol{k}_t + \boldsymbol{e}_t = \boldsymbol{H}_t \boldsymbol{x}_t + \boldsymbol{e}_t, \qquad \boldsymbol{H}_t = \boldsymbol{I}_V \otimes \boldsymbol{k}_t^{\mathsf{T}}, \qquad \boldsymbol{e}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_t). \tag{2}$$

Unless stated otherwise, $\boldsymbol{R}_t = \operatorname{diag}(r_{t,1}, \dots, r_{t,V}) \succ 0$. Diagonal observation noise preserves cross-value independence while allowing different value coordinates to have different reliability.

The two one-axis uncertainty families are

$$\underbrace{\boldsymbol{I}_V \otimes \boldsymbol{P}_t}_{\text{shared key-space uncertainty}} \qquad \text{and} \qquad \underbrace{\boldsymbol{B}_t \otimes \boldsymbol{I}_K}_{\text{value-wise isotropic uncertainty}}, \tag{3}$$

where $\boldsymbol{P}_t \in \mathbb{R}^{K\times K}$ and $\boldsymbol{B}_t = \operatorname{diag}(b_{t,1}, \dots, b_{t,V})$. The first is the family used by Diagonal KDN; the second is derived next.

2 Value-wise isotropic uncertainty

2.1 Retained family and state-space model

The value-wise isotropic family assigns one scalar key-space variance to every value coordinate:

$$\widetilde{\boldsymbol{s}}_{t,j} \mid \mathcal{F}_t \sim \mathcal{N}(\boldsymbol{s}_{t,j}, b_{t,j}\boldsymbol{I}_K), \qquad \operatorname{Cov}\Big(\operatorname{vec}(\widetilde{\boldsymbol{S}}_t) \mid \mathcal{F}_t\Big) = \boldsymbol{B}_t \otimes \boldsymbol{I}_K. \tag{4}$$

The columns remain conditionally independent, but their uncertainty need not be equal. Consider the columnwise process and observation model

$$\begin{aligned} &\widetilde{\boldsymbol{s}}_{t,j} = \boldsymbol{D}_t\widetilde{\boldsymbol{s}}_{t-1,j} + \boldsymbol{w}_{t,j}, \quad \boldsymbol{w}_{t,j} \sim \mathcal{N}(\boldsymbol{0}, \omega_{t,j}\boldsymbol{I}_K), \\ &v_{t,j} = \boldsymbol{k}_t^\top \widetilde{\boldsymbol{s}}_{t,j} + e_{t,j}, \qquad\quad e_{t,j} \sim \mathcal{N}(0, r_{t,j}), \qquad \boldsymbol{D}_t = \operatorname{diag}(\boldsymbol{\alpha}_t). \end{aligned} \tag{5}$$

The noises are independent across value coordinates and time. The current shared-noise model is recovered by tying $\omega_{t,j} = \omega_t$ and $r_{t,j} = r_t$.

The mean prediction is

$$\widehat{\boldsymbol{S}}_t = \boldsymbol{D}_t\boldsymbol{S}_{t-1}. \tag{6}$$

The channel-wise transition generally makes the exact predictive covariance anisotropic even when the retained covariance is isotropic. Define its average squared transition

$$a_t = \frac{1}{K}\operatorname{Tr}(\boldsymbol{D}_t\boldsymbol{D}_t^\top) = \frac{1}{K}\sum_{i=1}^{K}\alpha_{t,i}^2. \tag{7}$$

Projecting the predictive covariance to its average marginal variance gives

$$\widehat{b}_{t,j} = a_t b_{t-1,j} + \omega_{t,j}, \qquad \widehat{\boldsymbol{P}}_{t,j} = \widehat{b}_{t,j}\boldsymbol{I}_K. \tag{8}$$

This prediction is exact when $\boldsymbol{D}_t\boldsymbol{D}_t^\top = a_t\boldsymbol{I}_K$ and is otherwise the covariance-coordinate trace projection used by Isotropic KDN.

2.2 Exact conditional mean

Let

$$\delta_{t,j} = v_{t,j} - \boldsymbol{k}_t^\top\widehat{\boldsymbol{s}}_{t,j}. \tag{9}$$

The predictive cross-covariance and innovation variance are

$$\operatorname{Cov}(\widetilde{\boldsymbol{s}}_{t,j}, v_{t,j} \mid \mathcal{F}_{t-1}) = \widehat{b}_{t,j}\boldsymbol{k}_t, \qquad \Delta_{t,j} = r_{t,j} + \widehat{b}_{t,j}\|\boldsymbol{k}_t\|_2^2. \tag{10}$$

Therefore column j has scalar gain

$$\beta_{t,j} = \frac{\widehat{b}_{t,j}}{\Delta_{t,j}}, \qquad \boldsymbol{s}_{t,j} = \widehat{\boldsymbol{s}}_{t,j} + \beta_{t,j}\boldsymbol{k}_t\delta_{t,j}. \tag{11}$$

Every value column writes along the same key direction, but each column uses a different uncertainty-derived contraction. In matrix form, with $\boldsymbol{\beta}_t = (\beta_{t,1}, \dots, \beta_{t,V})^\top$,

$$\boldsymbol{S}_t = \widehat{\boldsymbol{S}}_t + \boldsymbol{k}_t\left[\boldsymbol{\beta}_t \odot \left(\boldsymbol{v}_t - \widehat{\boldsymbol{S}}_t^\top\boldsymbol{k}_t\right)\right]^\top. \tag{12}$$

Equivalently,

$$\boldsymbol{S}_t = \widehat{\boldsymbol{S}}_t - \boldsymbol{k}_t\boldsymbol{k}_t^\top\widehat{\boldsymbol{S}}_t\operatorname{diag}(\boldsymbol{\beta}_t) + \boldsymbol{k}_t(\boldsymbol{\beta}_t \odot \boldsymbol{v}_t)^\top. \tag{13}$$

2.3 Reverse-KL projection and information scan

The exact posterior precision for column j is

$$\boldsymbol{C}^{\star}_{t,j} = \widehat{c}_{t,j}\boldsymbol{I}_K + \frac{1}{r_{t,j}}\boldsymbol{k}_t\boldsymbol{k}_t^{\mathsf{T}}, \qquad \widehat{c}_{t,j} = \widehat{b}_{t,j}^{-1}. \tag{14}$$

It is dense for a dense key, so it must be projected before the next token.

Proposition 1 (Value-wise isotropic reverse-KL projection)**.** *Let $\pi_{t,j}$ be the exact Gaussian posterior defined by Equations (5)–(14). Over*

$$q_{t,j} = \mathcal{N}(\boldsymbol{m}_{t,j}, b_{t,j}\boldsymbol{I}_K),$$

the unique minimizer of $\mathrm{KL}(q_{t,j} \,\|\, \pi_{t,j})$ *is*

$$\boldsymbol{m}_{t,j} = \boldsymbol{s}_{t,j}, \qquad b_{t,j} = \frac{K}{\mathrm{Tr}(\boldsymbol{C}^{\star}_{t,j})} = \left(\widehat{c}_{t,j} + \frac{\|\boldsymbol{k}_t\|_2^2}{K r_{t,j}}\right)^{-1}. \tag{15}$$

Proof. For a Gaussian target with precision $\boldsymbol{C}^{\star}_{t,j}$, the covariance-dependent part of the reverse KL is $\frac{1}{2}\{b_{t,j}\,\mathrm{Tr}(\boldsymbol{C}^{\star}_{t,j}) - K\log b_{t,j}\}$. Its derivative vanishes only at $b_{t,j} = K/\,\mathrm{Tr}(\boldsymbol{C}^{\star}_{t,j})$. The mean-dependent quadratic is uniquely minimized by the exact posterior mean in Equation (11). □

Equation (15) is the literal variational update. As in Isotropic KDN, an information scale $\mu > 0$ can instead be introduced only in the retained post-write precision:

$$\widehat{c}_{t,j} = \frac{c_{t-1,j}}{a_t + \omega_{t,j}c_{t-1,j}}, \qquad u_{t,j} = \mu\frac{\|\boldsymbol{k}_t\|_2^2}{K r_{t,j}}, \qquad c_{t,j} = \widehat{c}_{t,j} + u_{t,j}. \tag{16}$$

Here $\mu = 1$ is the reverse-KL optimum. Values $\mu \neq 1$ calibrate future protection while leaving the current gain in Equation (11) unchanged.

Substitution gives a scalar Möbius recurrence for every value coordinate:

$$c_{t,j} = \frac{(1 + u_{t,j}\omega_{t,j})c_{t-1,j} + u_{t,j}a_t}{\omega_{t,j}c_{t-1,j} + a_t}, \qquad \boldsymbol{M}_{t,j} = \begin{bmatrix} 1 + u_{t,j}\omega_{t,j} & u_{t,j}a_t \\ \omega_{t,j} & a_t \end{bmatrix}. \tag{17}$$

Thus the uncertainty stage is V independent associative scalar scans.

2.4 When the model is genuinely different

If $b_{0,j}$, $\omega_{t,j}$, and $r_{t,j}$ are tied across j, then induction through Equations (8) and (16) keeps every $b_{t,j}$ and $\beta_{t,j}$ equal. Realized values cannot break this symmetry because Gaussian covariance updates do not depend on the realized observation. At least one value-wise initial variance, process uncertainty, or observation noise must vary for the model to differ from Isotropic KDN.

The sequential mean update still costs $O(KV)$, but the transition applied to column j is

$$\boldsymbol{A}_{t,j} = (\boldsymbol{I}_K - \beta_{t,j}\boldsymbol{k}_t\boldsymbol{k}_t^{\mathsf{T}})\boldsymbol{D}_t. \tag{18}$$

Because $\boldsymbol{A}_{t,j}$ varies with j, the single compact-WY system shared by all value columns in current KDN kernels no longer applies. An exact parallel implementation needs value-batched systems or a grouped-value approximation.

3 KDN-3: combined key–value factorized uncertainty

3.1 Separable diagonal family

KDN-3 combines the two one-axis families by retaining

$$\begin{aligned} \boldsymbol{P}_t &= \operatorname{diag}(p_{t,1},\ldots,p_{t,K}) \succ 0, & \boldsymbol{B}_t &= \operatorname{diag}(b_{t,1},\ldots,b_{t,V}) \succ 0, \\ \boldsymbol{\Sigma}_t &= \boldsymbol{B}_t \otimes \boldsymbol{P}_t = (\boldsymbol{B}_t \otimes \boldsymbol{I}_K)(\boldsymbol{I}_V \otimes \boldsymbol{P}_t), & \boldsymbol{J}_t &= \boldsymbol{\Sigma}_t^{-1} = \boldsymbol{G}_t \otimes \boldsymbol{C}_t, \end{aligned} \tag{19}$$

where $\boldsymbol{G}_t = \boldsymbol{B}_t^{-1}$ and $\boldsymbol{C}_t = \boldsymbol{P}_t^{-1}$. Hence the covariance and information forms are the same family: inversion simply inverts both diagonal factors. Entry (i,j) of the latent memory has variance

$$\operatorname{Var}(\widetilde{S}_{t,ij} \mid \mathcal{F}_t) = p_{t,i} b_{t,j}. \tag{20}$$

The model has $K+V-1$ covariance degrees of freedom, not KV, because

$$\boldsymbol{B}_t \otimes \boldsymbol{P}_t = (\lambda \boldsymbol{B}_t) \otimes (\boldsymbol{P}_t/\lambda) \qquad \text{for every } \lambda > 0. \tag{21}$$

We fix this internal factor gauge by imposing

$$\frac{1}{V}\sum_{j=1}^{V} \log b_{t,j} = 0 \qquad \Longleftrightarrow \qquad \det(\boldsymbol{B}_t) = 1, \tag{22}$$

and compensating $\boldsymbol{P}_t$ so that $\boldsymbol{B}_t \otimes \boldsymbol{P}_t$ is unchanged.

At the covariance-family level, the nested special cases are

$$\begin{array}{rcl} \boldsymbol{B}_t = \bar{b}_t \boldsymbol{I}_V,\ \boldsymbol{R}_t = r_t \boldsymbol{I}_V & \Longrightarrow & \text{Diagonal KDN}, \\ \boldsymbol{P}_t = \bar{p}_t \boldsymbol{I}_K & \Longrightarrow & \text{value-wise isotropic KDN}, \\ \boldsymbol{B}_t = \bar{b}_t \boldsymbol{I}_V,\ \boldsymbol{P}_t = \bar{p}_t \boldsymbol{I}_K,\ \boldsymbol{R}_t = r_t \boldsymbol{I}_V & \Longrightarrow & \text{Isotropic KDN}. \end{array} \tag{23}$$

The scalar factor in each row can be absorbed into the other covariance factor before applying the gauge in Equation (22). These are one-step covariance-family limits. Recovering the complete one-axis recurrences additionally requires constraining the corresponding factor and using its one-axis predictive projection; in particular, the channel-wise transition in Equation (25) does not preserve $\boldsymbol{P}_t \propto \boldsymbol{I}_K$ without the trace projection in Equation (8).

3.2 Factor-preserving prediction

The key-space mean transition remains

$$\widehat{\boldsymbol{S}}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1}, \qquad \boldsymbol{D}_t = \operatorname{diag}(\boldsymbol{\alpha}_t). \tag{24}$$

A simple retained prediction updates the two covariance factors by

$$\widehat{\boldsymbol{P}}_t = \boldsymbol{D}_t \boldsymbol{P}_{t-1} \boldsymbol{D}_t^{\mathsf{T}} + \boldsymbol{\Omega}_t^{(k)}, \qquad \widehat{\boldsymbol{B}}_t = \boldsymbol{B}_{t-1} + \boldsymbol{\Omega}_t^{(v)}, \tag{25}$$

with positive diagonal process factors. This equation should be read as a factor-preserving predictive parameterization. It corresponds to the positive-semidefinite full process covariance

$$\boldsymbol{Q}_t = \boldsymbol{B}_{t-1} \otimes \boldsymbol{\Omega}_t^{(k)} + \boldsymbol{\Omega}_t^{(v)} \otimes (\boldsymbol{D}_t \boldsymbol{P}_{t-1} \boldsymbol{D}_t^{\mathsf{T}}) + \boldsymbol{\Omega}_t^{(v)} \otimes \boldsymbol{\Omega}_t^{(k)}, \tag{26}$$

because adding Equation (26) to the transitioned prior gives $\widehat{\boldsymbol{B}}_t \otimes \widehat{\boldsymbol{P}}_t$. A generic additive separable process covariance would instead produce a sum of Kronecker products and leave the retained family.

After prediction, apply the gauge normalization in Equation (22). This normalization leaves the full predictive covariance and the full Kalman gain unchanged. It does rescale the factors used below: under $(\widehat{\boldsymbol{B}}_t, \widehat{\boldsymbol{P}}_t) \mapsto (\lambda \widehat{\boldsymbol{B}}_t, \widehat{\boldsymbol{P}}_t/\lambda)$, $\boldsymbol{g}_t \mapsto \boldsymbol{g}_t/\lambda$ and $\boldsymbol{\beta}_t \mapsto \lambda \boldsymbol{\beta}_t$, so their product is invariant.

3.3 Exact innovation and conditional mean

Define

$$\boldsymbol{g}_t = \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t, \qquad z_t = \boldsymbol{k}_t^\top \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t, \qquad \boldsymbol{\delta}_t = \boldsymbol{v}_t - \widehat{\boldsymbol{S}}_t^\top \boldsymbol{k}_t. \tag{27}$$

The innovation covariance and state–observation cross-covariance are

$$\boldsymbol{\Delta}_t = z_t \widehat{\boldsymbol{B}}_t + \boldsymbol{R}_t, \qquad \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{v}_t \mid \mathcal{F}_{t-1}) = \widehat{\boldsymbol{B}}_t \otimes \boldsymbol{g}_t. \tag{28}$$

The exact vectorized Kalman gain is therefore

$$\boldsymbol{K}_t = (\widehat{\boldsymbol{B}}_t \otimes \boldsymbol{g}_t) \boldsymbol{\Delta}_t^{-1}. \tag{29}$$

When $\boldsymbol{R}_t = \mathrm{diag}(r_{t,1}, \ldots, r_{t,V})$, define

$$\beta_{t,j} = \frac{\widehat{b}_{t,j}}{r_{t,j} + \widehat{b}_{t,j} z_t}, \qquad \boldsymbol{\beta}_t = (\beta_{t,1}, \ldots, \beta_{t,V})^\top. \tag{30}$$

The exact posterior mean has the rank-one matrix update

$$\boldsymbol{S}_t^\star = \widehat{\boldsymbol{S}}_t + \boldsymbol{g}_t (\boldsymbol{\beta}_t \odot \boldsymbol{\delta}_t)^\top. \tag{31}$$

This equation is the central behavioral generalization. The key factor $\widehat{\boldsymbol{P}}_t$ rotates and rescales the write direction through $\boldsymbol{g}_t$, while the value factor $\widehat{\boldsymbol{B}}_t$ produces relative write strengths across value coordinates. Their absolute scales are gauge-dependent; only $\boldsymbol{g}_t \beta_{t,j}$ and ratios among the $\beta_{t,j}$ are identifiable. Setting $\widehat{\boldsymbol{B}}_t = \boldsymbol{I}_V$ and $\boldsymbol{R}_t = r_t \boldsymbol{I}_V$ recovers the Diagonal-KDN gain; setting $\widehat{\boldsymbol{P}}_t = \boldsymbol{I}_K$ recovers Equation (12).

The one-step predictive negative log likelihood, up to the additive Gaussian constant, is

$$\mathcal{L}_{\mathrm{pred},t} = \frac{1}{2} \sum_{j=1}^{V} \left[\frac{\delta_{t,j}^2}{r_{t,j} + \widehat{b}_{t,j} z_t} + \log(r_{t,j} + \widehat{b}_{t,j} z_t) \right]. \tag{32}$$

Both the residual and its variance use the exclusive pre-write state.

3.4 Exact posterior and failure of Kronecker closure

The exact posterior covariance and precision are

$$\begin{aligned} \boldsymbol{\Sigma}_t^\star &= \widehat{\boldsymbol{B}}_t \otimes \widehat{\boldsymbol{P}}_t - (\widehat{\boldsymbol{B}}_t \boldsymbol{\Delta}_t^{-1} \widehat{\boldsymbol{B}}_t) \otimes (\boldsymbol{g}_t \boldsymbol{g}_t^\top), \\ \boldsymbol{J}_t^\star &= \widehat{\boldsymbol{B}}_t^{-1} \otimes \widehat{\boldsymbol{P}}_t^{-1} + \boldsymbol{R}_t^{-1} \otimes \boldsymbol{k}_t \boldsymbol{k}_t^\top. \end{aligned} \tag{33}$$

For diagonal $\boldsymbol{R}_t$, column j has covariance block

$$\boldsymbol{\Sigma}_{t,j}^\star = \widehat{b}_{t,j} \left(\widehat{\boldsymbol{P}}_t - \beta_{t,j} \boldsymbol{g}_t \boldsymbol{g}_t^\top \right). \tag{34}$$

Different $\beta_{t,j}$ values therefore produce differently shaped key-space covariance blocks, not merely differently scaled copies of one common matrix.

Proposition 2 (Exact separable closure)**.** *Assume $K > 1$, $\widehat{\boldsymbol{P}}_t \succ 0$, and $\boldsymbol{g}_t \boldsymbol{g}_t^\top \neq \boldsymbol{0}$. The blocks in Equation (34) can be written as $b_{t,j}^+ \boldsymbol{P}_t^+$ with one common, possibly dense, matrix $\boldsymbol{P}_t^+$ if and only if all $\beta_{t,j}$ are equal. For diagonal observation noise, this is equivalent to*

$$\frac{r_{t,j}}{\widehat{b}_{t,j}} = \rho_t \quad \textit{for every } j, \qquad \textit{or equivalently} \qquad \boldsymbol{R}_t = \rho_t \widehat{\boldsymbol{B}}_t. \tag{35}$$

Proof. If $\beta_{t,j} = \beta_t$, then $\boldsymbol{\Sigma}^\star_{t,j} = \widehat{b}_{t,j}(\widehat{\boldsymbol{P}}_t - \beta_t \boldsymbol{g}_t \boldsymbol{g}_t^\top)$, which is separable. Conversely, suppose two blocks with coefficients $\beta_a \neq \beta_b$ are proportional. Then for some $c > 0$,

$$\widehat{\boldsymbol{P}}_t - \beta_a \boldsymbol{g}_t \boldsymbol{g}_t^\top = c(\widehat{\boldsymbol{P}}_t - \beta_b \boldsymbol{g}_t \boldsymbol{g}_t^\top).$$

If $c \neq 1$, the left rearrangement contains the full-rank matrix $(1-c)\widehat{\boldsymbol{P}}_t$ while the other side has rank at most one, a contradiction. Thus $c = 1$ and $\beta_a = \beta_b$. Finally, $\beta_{t,j} = 1/(r_{t,j}/\widehat{b}_{t,j} + z_t)$, so equality of the gains is equivalent to Equation (35). □

The compatible choice $\boldsymbol{R}_t = \rho_t \widehat{\boldsymbol{B}}_t$ is mathematically closed in the value factor, but it makes that factor cancel from the gain:

$$\boldsymbol{S}^\star_t = \widehat{\boldsymbol{S}}_t + \frac{\boldsymbol{g}_t \boldsymbol{\delta}_t^\top}{\rho_t + z_t}. \tag{36}$$

The value uncertainty then has no effect on the posterior mean. Moreover, the common key-space posterior remains dense for a dense key, so the diagonal $\boldsymbol{P}_t$ restriction still requires the usual KDN projection. Useful value-dependent gains and exact Kronecker closure therefore cannot generally be obtained simultaneously.

3.5 Joint reverse-KL projection

We retain the exact mean in Equation (31) and project only its covariance. Let

$$q_t = \mathcal{N}(\mathrm{vec}(\boldsymbol{S}^\star_t), \boldsymbol{B}_t \otimes \boldsymbol{P}_t), \qquad \boldsymbol{B}_t = \mathrm{diag}(\boldsymbol{b}_t), \quad \boldsymbol{P}_t = \mathrm{diag}(\boldsymbol{p}_t). \tag{37}$$

Define four candidate-factor summaries

$$\begin{aligned} A_B &= \sum_{j=1}^{V} \frac{b_{t,j}}{\widehat{b}_{t,j}}, \quad T_B = \sum_{j=1}^{V} \frac{b_{t,j}}{r_{t,j}}, \\ A_P &= \sum_{i=1}^{K} \frac{p_{t,i}}{\widehat{p}_{t,i}}, \quad Z_P = \sum_{i=1}^{K} p_{t,i} k_{t,i}^2. \end{aligned} \tag{38}$$

Proposition 3 (Joint reverse-KL projection)**.** *Up to the factor gauge in Equation (21), the stationary point of* $\mathrm{KL}(q_t \,\|\, \pi^\star_t)$*, where* $\pi^\star_t$ *has covariance Equation (33), satisfies*

$$b_{t,j} = \frac{K}{A_P/\widehat{b}_{t,j} + Z_P/r_{t,j}}, \qquad p_{t,i} = \frac{V}{A_B/\widehat{p}_{t,i} + T_B k_{t,i}^2}. \tag{39}$$

Equivalently, with

$$L_{t,ij} = \frac{1}{\widehat{p}_{t,i}\widehat{b}_{t,j}} + \frac{k_{t,i}^2}{r_{t,j}}, \qquad W_{t,ij} = p_{t,i} b_{t,j} L_{t,ij}, \tag{40}$$

the positive matrix $\boldsymbol{W}_t$ *has row sums* V *and column sums* K*.*

Proof. Using the precision in Equation (33), the covariance-dependent part of twice the reverse KL is

$$A_B A_P + T_B Z_P - K \log\det \boldsymbol{B}_t - V \log\det \boldsymbol{P}_t + \mathrm{const}. \tag{41}$$

Differentiating with respect to $b_{t,j}$ and $p_{t,i}$ yields Equation (39). Equivalently, $\partial/\partial p_{t,i} = 0$ gives $\sum_j W_{t,ij} = V$, and $\partial/\partial b_{t,j} = 0$ gives $\sum_i W_{t,ij} = K$. Multiplying every $b_{t,j}$ by λ and dividing every $p_{t,i}$ by λ leaves both the objective and the full covariance unchanged, which is exactly the factor gauge. □

In log-factor coordinates, Equation (41) is convex and is strictly convex after the gauge in Equation (22) is fixed. The positive fixed point is therefore unique in that gauge. It is a matrix-scaling problem, and alternating updates

$$\begin{aligned} b_{t,j} &\leftarrow \frac{K}{A_P/\widehat{b}_{t,j} + Z_P/r_{t,j}}, \\ p_{t,i} &\leftarrow \frac{V}{A_B/\widehat{p}_{t,i} + T_B k_{t,i}^2}, \end{aligned} \tag{42}$$

followed by gauge normalization solve it without constructing a $K \times V$ matrix. These are the standard alternating matrix-scaling updates and converge to the unique gauge-equivalence class because $L_{t,ij} > 0$. Each sweep costs $O(K+V)$ because $L_{t,ij}$ is the sum of two rank-one terms, one strictly positive and one nonnegative. The projection is nevertheless coupled and iterative at every token. It is not two independent Möbius scans over time.

The one-sided boundaries recover the previous models. Holding $\boldsymbol{B}_t = \widehat{\boldsymbol{B}}_t$ fixed gives

$$p_{t,i} = \left[\widehat{p}_{t,i}^{-1} + \left(\frac{1}{V} \sum_{j=1}^{V} \frac{\widehat{b}_{t,j}}{r_{t,j}} \right) k_{t,i}^2 \right]^{-1}. \tag{43}$$

For $\widehat{\boldsymbol{B}}_t = \boldsymbol{I}_V$ and $\boldsymbol{R}_t = r_t \boldsymbol{I}_V$, this is the ordinary Diagonal-KDN reverse-KL update. Holding $\boldsymbol{P}_t = \widehat{\boldsymbol{P}}_t$ fixed gives

$$b_{t,j} = \left[\widehat{b}_{t,j}^{-1} + \frac{z_t}{K r_{t,j}} \right]^{-1}, \tag{44}$$

which is the value-wise isotropic update when $\widehat{\boldsymbol{P}}_t = \boldsymbol{I}_K$. These equations describe one posterior projection. Recovering the full value-wise isotropic recurrence also requires the trace-projected prediction in Equation (8). Applying both one-sided full-information updates independently would double-count the same observation; the coupled projection is what allocates the information between factors.

An optional protection scale may multiply the measurement term $k_{t,i}^2/r_{t,j}$ in Equation (40). If the same scale also tempers the likelihood and changes the current gain, it defines a different Gaussian observation model. If the exact gain in Equation (31) is kept fixed, the scaled projection is instead a retained future-protection calibration, as in the one-axis KDN variants; it is not the reverse-KL optimum for Equation (2).

3.6 A practical reference recurrence

A sequential KDN-3 reference implementation can use the following token step:

1. Predict $\widehat{\boldsymbol{S}}_t$, $\widehat{\boldsymbol{P}}_t$, and $\widehat{\boldsymbol{B}}_t$ with Equations (24) and (25).
2. Fix the internal factor gauge by Equation (22).
3. Compute $\boldsymbol{g}_t$, z_t, $\boldsymbol{\beta}_t$, and the exact posterior mean $\boldsymbol{S}_t^\star$ from Equations (27)–(31).
4. Initialize the covariance projection at the predictive factors and run a fixed number of alternating updates in Equation (42).
5. Reapply the gauge normalization and carry the projected factors to token $t+1$.

With N_{proj} alternating sweeps, uncertainty work is $O(N_{\mathrm{proj}}(K+V))$ per token and the memory update is $O(KV)$. This is suitable as a correctness oracle and as a small-scale experiment. It is not yet a drop-in replacement for the current scan-parallel KDN kernel.

4 Connection to Gated DeltaNet-2

4.1 The two updates side by side

Use the same predicted memory $\widehat{\boldsymbol{S}}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1}$. Gated DeltaNet-2 (GDN-2) can be written as

$$\begin{aligned} \boldsymbol{S}_t^{\mathrm{GDN2}} &= \widehat{\boldsymbol{S}}_t - \boldsymbol{k}_t(\boldsymbol{b}_t^{\mathrm{erase}} \odot \boldsymbol{k}_t)^\top \widehat{\boldsymbol{S}}_t + \boldsymbol{k}_t(\boldsymbol{w}_t \odot \boldsymbol{v}_t)^\top, \\ \boldsymbol{b}_t^{\mathrm{erase}} &\in \mathbb{R}^K, \qquad \boldsymbol{w}_t \in \mathbb{R}^V, \end{aligned} \tag{45}$$

where $\boldsymbol{b}_t^{\mathrm{erase}}$ is a key-channel erase gate and $\boldsymbol{w}_t$ is a value-channel write gate. Both are predicted directly from the current token and are sigmoid-valued in the default implementation; an optional GDN-2 variant expands only the erase-gate range. The keys are ℓ_2-normalized in both model families. Defining the pseudo-key and pseudo-value

$$\widetilde{\boldsymbol{k}}_t = \boldsymbol{b}_t^{\mathrm{erase}} \odot \boldsymbol{k}_t, \qquad \widetilde{\boldsymbol{v}}_t = \boldsymbol{w}_t \odot \boldsymbol{v}_t, \tag{46}$$

gives the residual-like form

$$\boldsymbol{S}_t^{\mathrm{GDN2}} = \widehat{\boldsymbol{S}}_t + \boldsymbol{k}_t \left(\widetilde{\boldsymbol{v}}_t - \widehat{\boldsymbol{S}}_t^\top \widetilde{\boldsymbol{k}}_t \right)^\top. \tag{47}$$

This resembles a Kalman residual update algebraically, but its erased read uses $\widetilde{\boldsymbol{k}}_t$ while its injected target is $\widetilde{\boldsymbol{v}}_t$. In general it is not the posterior mean for the original observation $\boldsymbol{v}_t = \widetilde{\boldsymbol{S}}_t^\top \boldsymbol{k}_t + \boldsymbol{e}_t^{\mathrm{obs}}$.

Expanding the KDN-3 mean update in Equation (31) gives

$$\boldsymbol{S}_t^{\mathrm{KDN3}} = \widehat{\boldsymbol{S}}_t - \boldsymbol{g}_t \boldsymbol{k}_t^\top \widehat{\boldsymbol{S}}_t \operatorname{diag}(\boldsymbol{\beta}_t) + \boldsymbol{g}_t(\boldsymbol{\beta}_t \odot \boldsymbol{v}_t)^\top, \qquad \boldsymbol{g}_t = \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t. \tag{48}$$

Both models therefore use an $O(KV)$ low-rank correction, and both can modulate the incoming value coordinatewise. Their factorizations differ:

Model	outer write direction	erased read direction	value multiplier
GDN-2	$\boldsymbol{k}_t$	$\boldsymbol{b}_t^{\mathrm{erase}} \odot \boldsymbol{k}_t$	$\boldsymbol{w}_t$
KDN-3	$\boldsymbol{g}_t = \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t$	$\boldsymbol{k}_t$	$\boldsymbol{\beta}_t$

For KDN-3, the same $\beta_{t,j}$ multiplies the observation write and the erasure of the predicted value in column j. This tie is forced by the Kalman innovation. GDN-2 instead predicts $\boldsymbol{b}_t^{\mathrm{erase}}$ and $\boldsymbol{w}_t$ independently, deliberately decoupling how much old content is erased from how much new value is injected.

The two covariance factors explain the two visible parts of the KDN-3 update. The key factor $\widehat{\boldsymbol{P}}_t$ changes the outer write direction from $\boldsymbol{k}_t$ to $\boldsymbol{g}_t$, while the value factor $\widehat{\boldsymbol{B}}_t$ and observation noise determine the relative entries of $\boldsymbol{\beta}_t$. These gates are functions of the carried uncertainty state and therefore depend on the history, not only on the current token representation.

4.2 Exact intersection of the update families

The similarity between Equations (45) and (48) does not imply that either family contains the other. Their common intersection is much smaller.

Proposition 4 (Algebraic GDN-2–KDN-3 intersection)**.** *At the level of the memory-update equations, assume a nondegenerate update with $\boldsymbol{k}_t \neq \boldsymbol{0}$ and require Equations (45) and (48) to agree for arbitrary $\widehat{\boldsymbol{S}}_t$ and $\boldsymbol{v}_t$. Then there must be scalars $\lambda_t > 0$ and $\beta_t > 0$ such that*

$$\begin{aligned} \boldsymbol{g}_t &= \lambda_t \boldsymbol{k}_t, & \boldsymbol{\beta}_t &= \beta_t \mathbf{1}_V, \\ \boldsymbol{w}_t &= \lambda_t \beta_t \mathbf{1}_V, & \boldsymbol{b}_t^{\mathrm{erase}} \odot \boldsymbol{k}_t &= \lambda_t \beta_t \boldsymbol{k}_t. \end{aligned} \tag{49}$$

Conversely, these conditions make the two updates identical.

Proof. Equality of the coefficients multiplying each arbitrary $v_{t,j}$ requires

$$\beta_{t,j}\boldsymbol{g}_t = w_{t,j}\boldsymbol{k}_t. \tag{50}$$

For a nonzero write this forces $\boldsymbol{g}_t = \lambda_t \boldsymbol{k}_t$ and $w_{t,j} = \lambda_t \beta_{t,j}$. Equality of the linear maps applied to each arbitrary memory column then requires

$$\lambda_t \beta_{t,j} \boldsymbol{k}_t \boldsymbol{k}_t^\top = \boldsymbol{k}_t (\boldsymbol{b}_t^{\text{erase}} \odot \boldsymbol{k}_t)^\top \qquad \text{for every } j. \tag{51}$$

The right-hand side is shared across value columns, so every $\beta_{t,j}$ must equal one scalar β_t, and $\boldsymbol{b}_t^{\text{erase}} \odot \boldsymbol{k}_t = \lambda_t \beta_t \boldsymbol{k}_t$. Substitution proves sufficiency. □

For the implemented default GDN-2 parameterization, the converse additionally requires $\lambda_t \beta_t \in (0, 1)$ so that both gates lie in their sigmoid ranges. The optional negative-eigenvalue variant permits an erase coefficient up to two but leaves the write gate in $(0, 1)$.

For a dense key and diagonal $\widehat{\boldsymbol{P}}_t$, the condition $\boldsymbol{g}_t = \lambda_t \boldsymbol{k}_t$ requires $\widehat{p}_{t,i} = \lambda_t$ on the support of $\boldsymbol{k}_t$. The intersection therefore reduces to the scalar-gated delta update

$$\boldsymbol{S}_t = \widehat{\boldsymbol{S}}_t + \gamma_t \boldsymbol{k}_t (\boldsymbol{v}_t - \widehat{\boldsymbol{S}}_t^\top \boldsymbol{k}_t)^\top, \qquad \gamma_t = \lambda_t \beta_t, \tag{52}$$

which is the KDA form. Away from this intersection, KDN-3 can rotate the write direction through key-space uncertainty, which GDN-2 cannot express with its fixed outer direction $\boldsymbol{k}_t$; GDN-2 can independently choose erase and write gates, which a coherent KDN-3 innovation cannot express.

The one-axis boundaries expose the two separate mismatches. When $\widehat{\boldsymbol{P}}_t \propto \boldsymbol{I}_K$ but $\boldsymbol{\beta}_t$ remains value dependent, KDN-3 reduces to the value-wise isotropic update: it shares GDN-2's outer direction and value-wise write scaling, but still uses the same $\beta_{t,j}$ for both halves of each column's innovation. When $\widehat{\boldsymbol{B}}_t \propto \boldsymbol{I}_V$ and $\boldsymbol{R}_t \propto \boldsymbol{I}_V$, KDN-3 reduces to Diagonal KDN: its value gate becomes common, but the anisotropic direction $\widehat{\boldsymbol{P}}_t \boldsymbol{k}_t$ generally cannot be represented by GDN-2.

4.3 Interpretation and controlled comparisons

GDN-2 may be viewed as a discriminative relaxation of an uncertainty-gated delta rule. It predicts the two sides of the update directly and gains token-local flexibility, but it does not carry a covariance whose predictive variance or posterior contraction constrains those gates. KDN-3 instead derives its update from the separable predictive covariance and diagonal observation noise. Its restrictions provide a calibrated predictive distribution and a causal notion of future protection, at the cost of the coupled covariance projection derived above.

The connection suggests four controlled ablations:

1. **Native scalar innovation tying.** Within the existing GDN-2 operator, constrain $\boldsymbol{b}_t^{\text{erase}} = \gamma_t \mathbf{1}_K$ and $\boldsymbol{w}_t = \gamma_t \mathbf{1}_V$. This tests the benefit of decoupling while retaining the native shared erase system.

2. **Value-wise tied innovation.** Replace the shared erase operator by $-\boldsymbol{k}_t \boldsymbol{k}_t^\top \widehat{\boldsymbol{S}}_t \operatorname{diag}(\boldsymbol{\beta}_t)$ and use the same $\boldsymbol{\beta}_t$ on the incoming value. This is a structural recurrence ablation, not merely a gate constraint on GDN-2.

3. **Gain direction.** Replace $\boldsymbol{g}_t = \widehat{\boldsymbol{P}}_t \boldsymbol{k}_t$ by a collinear direction

$$\boldsymbol{g}_{t,\parallel} = \frac{\boldsymbol{k}_t^\top \boldsymbol{g}_t}{\|\boldsymbol{k}_t\|_2^2} \boldsymbol{k}_t,$$

   which preserves $\boldsymbol{k}_t^\top \boldsymbol{g}_t$ and hence the current-key contraction while removing the off-key component.

4. **Stateful versus direct gates.** Match instantaneous write strengths while comparing uncertainty-derived gates with gates predicted only from the current token.

Useful diagnostics are the cosine between $\boldsymbol{g}_t$ and $\boldsymbol{k}_t$, dispersion across $\beta_{t,j}$, disagreement between GDN-2 erase and write strengths, predictive innovation NLL, and repeated-key protection.

5 Parallelism and approximation choices

The value-wise model in Section 2 has independent scalar information recurrences, but its value-dependent gains require separate memory systems. KDN-3 adds a second difficulty: its covariance projection couples all key and value factors at every token. Three implementation levels should be distinguished:

1. **Sequential joint projection.** Run the exact gain and several alternating reverse-KL sweeps per token. This is the reference semantics.
2. **Grouped values.** Share one b and one gain across a small value group. This reduces the number of memory systems while retaining coarse value reliability.
3. **Chunk-frozen factors.** Freeze or lag one factor within a chunk, use the corresponding one-sided update, and commit a joint projection at the boundary. This may recover useful matrix operations, but it is an approximation whose error must be measured against the sequential reference.

The compatible-noise construction in Equation (35) restores a shared gain, but it also removes the value factor from the mean and is therefore a control rather than a full KDN-3 implementation.

6 Parameterization, gauges, and validation

Positive factors. Let $\boldsymbol{h}_t$ denote the observed token representation supplied to the uncertainty heads. Predict diagonal process increments and observation noise through positive floors, for example

$$\boldsymbol{\omega}_t^{(k)} = \omega_{k,\min} + \mathrm{softplus}(f_k(\boldsymbol{h}_t)), \quad \boldsymbol{\omega}_t^{(v)} = \omega_{v,\min} + \mathrm{softplus}(f_v(\boldsymbol{h}_t)), \quad \boldsymbol{r}_t = r_{\min} + \mathrm{softplus}(f_r(\boldsymbol{h}_t)). \tag{53}$$

Fixing $\boldsymbol{r}_t = r\mathbf{1}_V$ while learning value-specific $\boldsymbol{\omega}_t^{(v)}$ is a useful first experiment: it breaks value symmetry without learning the most direct $b_{t,j}/r_{t,j}$ confound. In addition to the exact internal factor gauge in Equation (21), jointly scaling the full covariance and observation noise produces the usual Kalman covariance-scale gauge.

Symmetry breaking. Initialize $\boldsymbol{B}_0 = \boldsymbol{I}_V$ and $\boldsymbol{P}_0 = \boldsymbol{I}_K$ for a neutral gauge. Value uncertainty remains tied unless the initial value factor, value-process uncertainty, or observation noise varies across value coordinates. Realized values alone cannot change covariance symmetry.

Required tests. A KDN-3 implementation should pass the following checks before language-model training:

1. Compare the exact mean and covariance in Equations (31) and (33) against a dense Kalman update on small random systems.
2. Verify the Diagonal-KDN and value-wise isotropic limits in Equations (43) and (44).
3. Check factor-gauge invariance numerically under $(\boldsymbol{B}, \boldsymbol{P}) \mapsto (\lambda \boldsymbol{B}, \boldsymbol{P}/\lambda)$.

4. Confirm that the alternating projection satisfies the row- and column-sum conditions in Equation (40) and decreases the reverse-KL objective.
5. Compare zero, one, two, and converged projection sweeps; report covariance error, gain error at later tokens, throughput, and peak memory.
6. Test repeated keys, orthogonal keys with identical elementwise squares, value-specific corruption, and value-basis rotations. The last test exposes the basis dependence of a diagonal value factor.

7 Limitations and experimental contract

KDN-3 represents a rank-one factorization of the $K \times V$ table of entrywise variances. It is strictly richer than either one-axis family but much smaller than a fully independent diagonal covariance with KV states. It still assumes no cross-key or cross-value posterior covariance, depends on the learned key and value bases, and introduces both an exact factor gauge and strong confounding between value variance and observation noise.

The most important limitation is computational rather than notational. The exact posterior mean is inexpensive, but the joint covariance projection is coupled and the value-dependent gains break the shared memory transition. A claimed KDN-3 kernel must therefore specify which projection it implements, how many alternating sweeps it uses, how the gauge is fixed, and whether the memory system is value-wise, grouped, or chunk-approximated. Results should be compared with the two one-sided boundaries at matched parameter count and effective write strength.

The decisive initial experiment is a small synthetic ablation with four arms:

$$\begin{array}{ll} \text{Diagonal KDN,} & \text{value-wise isotropic KDN,} \\ \text{KDN-3 with one projection sweep,} & \text{KDN-3 with a converged projection.} \end{array} \tag{54}$$

This separates the benefit of two-axis uncertainty from the cost and approximation error of the joint projection before committing to a production kernel.

8 Conclusion

Value-wise isotropic uncertainty and Diagonal KDN allocate the same kind of scalar uncertainty budget to opposite axes of the memory matrix. KDN-3 combines them through the separable covariance $\boldsymbol{B}_t \otimes \boldsymbol{P}_t$. Its exact mean update cleanly factors into a key-wise uncertainty direction and value-wise uncertainty gates. Its covariance update does not factor cleanly: useful value-dependent gains destroy Kronecker closure, and the principled reverse-KL projection is a coupled matrix-balancing problem. This distinction is the central design constraint. KDN-3 is mathematically well defined as a sequential filter and a useful research model, but a scan-parallel approximation must be stated and validated separately.